Insurance Claim Approval Prediction Using Machine Learning Classification

ABSTRACT

Insurance claim approval is an important process in the insurance industry, where companies evaluate claims submitted by policyholders and decide whether to approve or reject them. Traditional manual assessment can be time-consuming and may lead to inconsistencies. Machine learning provides an effective approach to analyzing historical insurance data and supporting faster, more consistent claim evaluation.

The main objective of this project is to develop a classification-based machine learning system for predicting insurance claim approval using structured data. The dataset contains features such as policyholder age, annual income, policy type, policy tenure, premium amount, claim amount, previous claims count, employment status, documentation completeness, payment history, claim reason, and fraud flag. The claim identifier is excluded from model training because it is only used to identify individual records. Data preprocessing techniques, including missing-value handling where required, categorical-variable encoding, and feature scaling where appropriate, are applied before training.

Three supervised machine learning algorithms—K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Logistic Regression—are implemented to classify claims as approved or rejected. The dataset is divided into training and testing sets to evaluate model performance. Accuracy, precision, recall, F1-score, and a confusion matrix are used to compare the models and identify a suitable classifier.

The proposed system demonstrates the practical application of machine learning in insurance analytics. It is an educational decision-support prototype intended to support preliminary claim assessment and does not replace professional review or the insurer's final decision.

KEYWORDS

Insurance,ClaimApproval,MachineLearning,Classification,K-NearestNeighbors (KNN),SupportVectorMachine(SVM),LogisticRegression,Prediction,SupervisedLearning,DataPreprocessing,FeatureScaling,OneHotEncoding,Accuracy,Precision,Recall

PROBLEM STATEMENT

Insurance providers need to assess many claims using information about policyholders, policies, submitted amounts, supporting documentation, and payment history. Manual review can take time and decisions may not always be consistent. This project addresses the classification problem of predicting whether an insurance claim is approved or rejected from historical structured data. The model is designed for learning and preliminary decision support, while the final decision remains with authorized insurance professionals.

OBJECTIVE

The primary objective is to build and evaluate a machine learning classification model for insurance claim approval prediction. The project aims to prepare the dataset, encode categorical variables, scale features when needed, train KNN, SVM, and Logistic Regression models, and compare their performance using standard classification metrics. The target column is Claim_Approved, where 1 represents an approved claim and 0 represents a rejected claim.

EXISTING SYSTEM

In a conventional claim assessment process, claim details and supporting documents are reviewed by insurance staff according to the insurer's procedures and policy rules. Although professional review is essential, a manual-first workflow may require substantial time and effort, particularly when the number of claims is high. A data-driven model can be studied as an additional tool for identifying patterns in historical records.

DISADVANTAGES OF EXISTING SYSTEM

- Claim assessment may take considerable time when many applications are received.
- Similar cases may be assessed inconsistently if review practices differ.
- It can be difficult to identify patterns across large volumes of historical claim records manually.
- A manual-only workflow offers limited automated comparison of alternative classification models.

PROPOSED SYSTEM

The proposed approach uses supervised machine learning to predict the Claim_Approved target. First, the dataset is inspected for data types, missing values, and duplicate records. Claim_ID is removed from the predictor set because it is an identifier rather than a meaningful claim characteristic.Categorical Columns—such as Policy_Type, Employment_Status, Documentation_Complete, Payment_History, Claim_Reason, and Fraud_Flag—are encoded into numeric form. Feature scaling is applied where appropriate, particularly for distance- and marginbased models. The data is then divided into training and testing sets. KNN, SVM, and Logistic Regression are trained and evaluated with accuracy, precision, recall, F1-score, and a confusion matrix.

The model with suitable test performance can be selected for demonstration.

ADVANTAGES OF PROPOSED SYSTEM

- Provides a structured way to study insurance claim approval patterns.
- Allows comparison of three supervised classification algorithms.
- Uses common evaluation metrics to make model performance easier to interpret.
- Can support learning and preliminary assessment while keeping human review in the decision process.

DATASET DETAILS

The supplied educational dataset contains 600 claim records and 14 columns. The target column is **Claim_Approved**: 1 means Approved and 0 means Rejected. The dataset includes policyholder and claim information such as Age, Annual_Income, Policy_Type, Policy_Tenure_Years, Premium_Amount,Claim_Amount,Previous_Claims_Count,Employment_Status, Documentation_Complete, Payment_History, Claim_Reason, and Fraud_Flag. Claim_ID is a unique identifier and should be excluded from model training.

EXPECTED OUTCOME

The project is expected to produce trained KNN, SVM, and Logistic Regression classifiers and a comparison of their test-set performance. Actual performance values should be reported only after running the experiments on

the dataset. Since the dataset is educational and synthetic, the model should not be treated as a real-world insurance decision system without validation, fairness checks, privacy safeguards, and expert oversight.

REFERENCES

1. Scikit-learn documentation, “Supervised learning” and “Model evaluation.”https://scikitlearn.org/stable/supervised_learning.html
2. Scikit-learn documentation, “Preprocessing data.” https://scikitlearn.org/stable/modules/preprocessing.html
3. Scikit-learn documentation, “Classification metrics.” https://scikitlearn.org/stable/modules/model_evaluation.html